Quantum Sentinel: Resolving the Protection Paradox in Spacetime Quantum Error Detection via an Open-Source Interactive Gamified Platform

Abstract

Quantum computers are sensitive to noise, and error-detection checks offer a way to identify unreliable executions before their outcomes are used. These checks, however, require additional qubits and gates, while post-selection discards runs in which an error is detected. Consequently, improving the quality of retained results does not necessarily improve the amount of useful information obtained from a fixed experimental budget. Here we examine this trade-off for spacetime Pauli checks in simulated Clifford circuits, with unprotected Clifford-plus-T circuits used to probe the limits of the comparison. We evaluate conditional output-distribution fidelity, the fraction of executions retained and a clearly defined composite utility that weights fidelity by retention. The simulation benchmark comprises 4,944 circuit-configuration records spanning circuit widths, depths and three synthetic two-qubit noise regimes. A representative comparison reports an increase in conditional fidelity from 0.964 to 0.990, corresponding to a 3.60-fold reduction in the selected distribution-error measure. Nevertheless, the reported best composite-utility ratio was 0.967 (95% confidence interval, 0.960-0.973), and no evaluated protected configuration had a reported 95% confidence interval whose lower bound exceeded parity with the unprotected reference. Held-out strategy-selection comparisons similarly failed to demonstrate a gain over using no checks. These observations, restricted to the reported simulations and utility definition, show why quantum protection must be assessed jointly in terms of accuracy, accepted samples and implementation cost. A four-mission research interface combines recorded and live Qiskit runs, resource-use displays, staged syndrome filtering and a circuit inspector, making this trade-off visible without equating output-distribution agreement with full quantum-state fidelity.

Introduction

A quantum computer operates by manipulating qubits, physical systems that can carry information in combinations of quantum states. Unlike a conventional digital computation, the

information represented by a qubit can be disturbed by imperfect control, unwanted interactions with its environment and errors during measurement. As a circuit contains more operations, these imperfections accumulate. Error correction offers a principled route to reliable large-scale quantum computing but can require substantial additional hardware. On processors with more limited resources, error mitigation and error detection provide intermediate approaches to improving the reliability of computed results [1-4]. Foundational quantum error-correcting schemes were established by Shor and Steane [13,14]. Subsequent surface-code experiments demonstrated logical-error suppression and operation below the surface-code threshold [15,16].

Error detection asks a relatively simple question: *Did something happen during this execution that makes its result suspicious?* A detector introduces extra measurements whose outcomes, called syndromes, indicate whether a class of errors was observed. One can discard results with a non-trivial syndrome and analyse only the remaining executions. This process, known as post-selection, can increase the accuracy of the retained sample, but it does not repair the rejected computation. If a detector accepts only half of the runs, twice as many executions are needed on average to obtain the same number of accepted measurements, before accounting for any other overhead.

Coherent Pauli checks exploit predictable transformations of quantum operators in Clifford circuits, which comprise a restricted but important family of quantum operations [5,6]. Spacetime codes extend this concept by distributing compatible checks over the circuit's qubits and time steps. Their design can reduce the entangling-gate and connectivity costs of error detection compared with checks concentrated at a single location [7,8]. These advances establish that low-overhead error detection is possible; they do not imply that every additional check is beneficial for every scientific objective.

The unresolved *decision problem* addressed here is therefore operational rather than merely technical: when does additional detection improve the useful outcome of a noisy quantum computation, once both rejected measurements and protective operations are accounted for? A higher conditional fidelity alone cannot answer this question. Conversely, retaining every run maximizes the sample count without guaranteeing that the resulting distribution is accurate. An

informative evaluation must measure both dimensions and report the hardware overhead of the chosen checks.

For a metered quantum-computing service, the number of attempts needed to obtain an adequate accepted sample may also affect execution time and experimental expense. A fixed-shot comparison therefore captures a practically relevant component of resource use, although the composite fidelity-retention objective evaluated here is not a measured monetary cost function; provider billing structures and hardware access overheads would need separate analysis.

We study this question using synthetic-noise simulation of circuits protected by automatically generated spacetime Pauli checks. We also consider Clifford circuits modified by T gates, which supply a standard non-Clifford operation. These additional circuits test whether trends in output-distribution fidelity can be interpreted consistently when the ideal output itself changes. Our primary contribution is an explicit, reproducible evaluation formulation that is paired unprotected and protected results, an exact accounting of retained shots, a bounded resource-constrained strategy comparison, and held-out testing of proposed selection policies. We do not claim to introduce spacetime Pauli checks, prove a universal advantage of post-selection, or establish a new learning algorithm that beats established search methods.

Methods

Circuit construction and ideal reference distributions

For each tested seed, a base quantum circuit was defined by the number of data qubits and a depth parameter controlling layers of gates. Clifford payloads contain only operations whose conjugation action maps Pauli operators to Pauli operators. These circuits admit efficient stabilizer simulation, although exact statevector simulation is also practical for the small systems examined here. For the non-Clifford family, T gates were inserted into a related payload and the ideal computational-basis distribution was obtained from the pure-state output of the corresponding noiseless circuit. The evaluated family uses T counts of 0, 1, 2 and 4. The exact circuit-generator source, gate-level schedules and ideal reference arrays must be deposited for reproducibility; their full definitions must be included with the reproducibility release.

Synthetic noise model

The noise was applied to entangling CX and CZ operations, with symmetric independent readout flips, and that single-qubit operations were idealized. For clarity, we write the standard two-qubit depolarizing channel as

$$\mathrm{E}_2(\rho)=(1\text{-}p_2)\rho+p_2\frac{I_4}{4}$$

where p_2 is the depolarizing-mixture parameter and I_4 is the identity operator on two qubits. In Qiskit Aer, the parameterization and exact gate-noise attachment must be recorded explicitly to avoid confusing a depolarizing mixture probability with the probability of a particular non-identity Pauli error. The benchmark specifies $p_2\in\{0.01,0.04,0.08\}$ and a readout flip probability $p_{\mathrm{ro}}=0.01$. The implemented channel specification is synthetic and should not be described as a calibrated hardware noise model without additional evidence. Measurement-error mitigation for larger quantum processors has been investigated separately [17].

Spacetime check synthesis

For Clifford payloads, candidate spacetime Pauli checks were synthesized with qiskit-paulice, which constructs ancillary operations intended to leave the logical payload unchanged in the absence of faults [7,8]. Candidate policies varied the number of checks and an ordered subset of target data qubits. Each generated circuit was compared with the same payload without checks. The space of policies was bounded and did not enumerate all Pauli-check placements; neither the number of sampled target orders nor the observed best candidate guarantees a global optimum. The circuit overhead should be recorded as additional ancillary qubits, two-qubit gates and depth following a fixed, documented compilation procedure.

Syndrome filtering and accepted output

Let $C_s(x,z)$ be the number of attempted executions that yield payload bitstring x and syndrome register z. Accepted shots have the designated trivial syndrome z=0. Their conditional output distribution is

$$P_s(x|A)=\frac{C_s(x,0)}{\sum_{x'} C_s(x',0)}$$

provided the denominator is positive. If no shots are accepted, conditional fidelity is undefined and the composite utility is set to zero by convention. The accepted-shot fraction is computed with equation (2). The actual syndrome-bit convention should be verified against qiskit-paulice and the transpiled classical-register mapping, rather than inferred from printed bitstring order.

Statistical comparisons and selection bias

The circuit-generation seed, not an individual shot, is the intended replication unit for comparing strategy performance across independent circuit instances. A paired comparison should evaluate protected and unprotected circuits generated from the same seed and under the same physical noise settings. Acceptance uncertainty can be described with a binomial confidence interval; uncertainty in fidelity or utility across circuit families requires circuit-level resampling or an explicit hierarchical model. Selection policies must be trained on separate circuit instances from those used for their final evaluation. Where several hypotheses or strategies are tested, multiple comparisons and search-induced optimism should be accounted for. Verification of the reported intervals and held-out assessment requires public access to the original records, split definition and analysis code.

Software implementation

The computational implementation uses a Python simulation backend based on Qiskit Aer and qiskit-paulice, together with an interactive React interface for configuring candidate strategies and inspecting outcomes. Precomputed results allow exact-match retrieval of already evaluated configurations, whereas new configurations require an actual simulation. The software must not approximate unevaluated strategies by silently substituting neighboring cached configurations.

The software specification identifies a FastAPI service (api/server.py), Qiskit 2.5.2 [20], Qiskit Aer 0.17.2 and a React 18/TypeScript client built with Vite. The described service distinguishes circuit execution (/api/execute), availability monitoring (/api/health), mission definitions (/api/sectors) and recorded-benchmark queries (/api/benchmark). The benchmark dataset is

identified as results/runs.csv; the software versions and file provenance remain subject to the reproducibility checks specified under Code availability.

Results

A resource-aware measure of the protection trade-off

We first define the quantities needed to compare strategies fairly. Let $P_{\mathrm{ideal}}(x)$ denote the ideal probability of observing the bitstring x, and let $P_0(x)$ denote the measured distribution of the noisy, unprotected circuit. For a protected circuit, $P_s(x|A)$ is the output distribution conditioned on the event A that all selected syndrome checks are accepted. We quantify their similarity to the ideal target using the **squared Bhattacharyya overlap**:

$$F(P,Q)=\left(\sum_{x\in\{0,1\}^n}\sqrt{P(x)Q(x)}\right)^2$$

This quantity equals one when the observed and ideal *measurement distributions* agree. It is not the fidelity between complete quantum states: two different quantum states may yield the same measurement probabilities in a single basis. Throughout, $F_0=F(P_{\mathrm{ideal}},P_0)$ denotes the unprotected reference, and $F_s=F\left(P_{\mathrm{ideal}},P_s(\cdot|A)\right)$ denotes the conditional output-distribution fidelity after post-selection.

The other essential quantity is the accepted-shot fraction:

$$p_s=\frac{N_{\mathrm{accepted},s}}{N_{\mathrm{attempted},s}}.$$

We define a transparent **composite utility**, not a physical count of correct shots, as

$$U_s=p_sF_s,\qquad Y_s=\frac{U_s}{F_0}$$

Here $Y_s>1$ means that protection improves this *particular* composite objective over the baseline; it does not mean that protection necessarily lowers the cost of estimating every observable, improves full quantum-state fidelity, or saves cloud-computing expenditure. Other scientific tasks can require different decision functions.

The exact break-even condition is

$$p_s F_s > F_0 \quad \Leftrightarrow \quad F_s > \frac{F_0}{p_s}$$

Since $F_s \leq 1$, a necessary condition for any protected strategy to win under equation (3) is

$$p_s > F_0$$

This simple bound explains why post-selection can become especially expensive when the unprotected circuit is already highly accurate. For example, at $F_0=0.964$, a strategy retaining at most 96.4% of shots cannot improve U_s even if the accepted output becomes perfect. This is an algebraic property of the defined objective, not a new quantum threshold theorem.

Simulated circuit families and experimental coverage

The recorded simulation benchmark comprises 4,944 configuration-level records and 618 circuit groups. It specifies three data-register widths (n=4,5,6), depth settings of 3, 6 and 10, two-qubit depolarizing parameters of 0.01, 0.04 and 0.08, a readout bit-flip probability of 0.01, and eight independent circuit-generation seeds (19, 23, 41, 53, 71, 73, 89 and 101), with 2,048 attempted shots per sampled configuration as specified in the benchmark report. For protected Clifford payloads, between one and three automatically synthesized checks were explored across bounded target-qubit orders. The comparison family used T-gate counts of 0, 1, 2 and 4; the non-Clifford variants were evaluated without automatically inserted spacetime checks.

These settings define a structured simulation benchmark rather than an exhaustive search over all physical codes or hardware layouts. In particular, varying a target-qubit order does not enumerate every physically distinct Pauli check, and the reported number of rows should not be interpreted as the number of statistically independent circuit instances. The associated per-run counts, circuit descriptions, statistical scripts and provenance logs are essential source data and must accompany the final publication record. The numerical summaries in this manuscript should be rechecked against those files before submission.

Post-selection improves the reported accuracy of accepted outcomes

A representative Clifford-circuit comparison at a two-qubit noise parameter of 0.01 and depth setting 10. In that example, unprotected output-distribution fidelity was F_0=0.964, whereas a selected protected configuration achieved conditional fidelity F_s=0.990. The corresponding reduction in the chosen error quantity, 1-F, is

$$R_{\mathrm{err}}=\frac{1-F_0}{1-F_s}=\frac{0.036}{0.010}=3.60$$

This is a 3.60-fold reduction in the distribution-based infidelity measure for that reported comparison. It must not be described as a 3.60-fold improvement in quantum-state fidelity, nor is it evidence that all error mechanisms have been detected. The rounded values used in this example support the stated 3.60-fold calculation; any larger maximum requires its own explicitly reported pair of underlying fidelities.

Across the evaluated experimental conditions, the largest conditional fidelities summarized by depth and noise regime range from 0.889 to 0.990 (Table 1). These are maxima across tested candidates, not values tracked along one fixed circuit while only its depth changes. Their non-monotonic ordering therefore should not be interpreted as evidence that increasing circuit depth directly improves detection performance.

Two-qubit depolarizing parameter	**Depth 3**	**Depth 6**	**Depth 10**
0.01	0.987	0.982	0.990
0.04	0.965	0.943	0.955
0.08	0.930	0.889	0.902

Table 1. Maximum recorded conditional output-distribution fidelity by noise setting and depth. Each cell is a maximum over evaluated configurations, not a matched longitudinal comparison or an independent estimate of hardware performance.

Higher conditional fidelity did not yield a reported composite-utility gain

The benchmark summary reports a highest composite-utility ratio of Y_s=0.967 with a stated 95% confidence interval from 0.960 to 0.973. Among 510 summarized protected configurations, none had a 95% confidence interval whose lower limit exceeded Y_s=1. These comparisons do not establish any composite-utility advantage over the unprotected reference in the evaluated settings. The stronger claim that *no strategy can ever improve throughput* would be unjustified. The benchmark description locates this reported maximum at four data qubits, depth 3, a two-qubit noise parameter of 0.01 and one added check.

This is a negative result for one explicitly defined objective, not a failure of error detection. Conditional fidelity and retained-shot fraction reward different aspects of a calculation. A user who requires a minimum fidelity for an accepted result may rationally choose checks even when equation (3) prefers the unprotected circuit. Conversely, a user who wishes to maximize the composite score might choose no checks. The correct policy is therefore task-dependent.

Because the 510 reported summaries were selected from a bounded search, the stated maximum could also be affected by searching many candidates on the same data. Confidence intervals need to reflect the independence of circuit instances rather than treating every shot or strategy as an independent experimental replicate. Without the underlying dataset and exact interval-construction code, neither the stated confidence bounds nor their multiple-comparison properties can be independently verified here.

Non-Clifford gates reveal an important limitation of the metric

A T gate introduces a phase rotation $T=\mathrm{diag}(1,e^{i\pi/4})$ and does not, in general, preserve Pauli operators under conjugation. For example,

$$TXT^{\dagger} = \frac{X+Y}{\sqrt{2}}$$

As a result, automatic generation of Clifford Pauli checks cannot simply be transferred unchanged to arbitrary T-containing circuits. The unprotected Clifford-plus-T circuits therefore serve as a diagnostic comparison family, not a demonstration of spacetime-check performance across T gates. Existing work has examined valid checks in more general circuits, but the present benchmark does not evaluate such constructions [7]. The role of non-Clifford resources in universal quantum computation is established by the magic-state framework [18].

A representative comparison shows a change in baseline distribution fidelity from approximately 0.901 for a Clifford instance to approximately 0.938 for a four-T variant. The difference is about 0.037 from these rounded endpoints; the numerical analysis records an unrounded shift of 0.0375. This observation must not be attributed to the T gate reducing physical errors. Introducing T gates changes the intended output distribution itself, and the squared Bhattacharyya measure is sensitive to the shape of that reference distribution. Comparisons across T counts should therefore pair circuits carefully, track their distinct ideal references and avoid assigning a causal interpretation to the raw fidelity difference without further controls. Changes in how the ideal distribution concentrates probability mass can alter this classical overlap independently of any reduction in the physical noise channel.

Held-out strategy selection did not outperform the no-check reference

We additionally compare several policies for choosing protection strategies using circuit instances held apart from the instances used for selection. Each policy is evaluated on actually simulated candidates rather than on an interpolated score. The reported mean differences in composite-utility ratio relative to the unprotected reference were negative for all four active protection strategies (Table 2). These values support the limited interpretation that none of these policies demonstrated a positive mean advantage on the specified held-out comparison. They do not demonstrate an advantage for a machine-learning method, nor by themselves establish that the policies overfit: uncertainty estimates, the complete split definition and a common evaluation count are needed.

Selection approach	Mean change relative to baseline	Held-out groups
First available check	−0.1391	26
Maximum training-set conditional fidelity	−0.2104	26
Maximum training-set composite utility	−0.1437	26
Risk-aware utility rule (mean minus 1 standard error)	−0.1428	26
No-check reference	0.0000	27

Table 2. Held-out strategy-selection comparison. Differences are measured relative to the unprotected reference in the composite-utility ratio. No uncertainty estimates were supplied for these entries; the counts of held-out groups differ.

Together these results argue against equating a sophisticated strategy selector with better scientific performance. A learning-based approach should only be claimed as an advance if it improves a pre-specified, budget-matched endpoint on independent tests, with uncertainty estimates that support the comparison.

Interactive gaming platform for investigating quantum protection decisions

To connect the resource-allocation problem with decisions that can be inspected and reproduced, we developed Quantum Sentinel as an interactive gaming platform accompanying the simulations. The interface presents an error-prone quantum calculation as a mission to be protected. This framing is intended for readers with no background in quantum information: a data qubit carries the calculation, an ancillary qubit serves as an additional error sensor, and a syndrome is the measurement that indicates whether a particular sensor has detected a fault. The visual mission map is a navigation metaphor, not a representation of a physical quantum state.

The client organises these research tasks as four mission sectors: Signal Rescue, Resource Siege, Benchmark Duel and T-Gate Frontier. A mission map introduces the circuit setting and scientific target before the participant enters the strategy workbench. The map's three-

dimensional celestial visualisation, procedural audio and experience-point system are presentation devices rather than representations of the simulated quantum state or independently validated measures of learning.

The platform links a React and TypeScript client to a Python FastAPI service and a Qiskit Aer simulation workflow. A participant chooses a circuit instance, the ordered data qubits on which to request checks, the number of checks, and limits on ancilla qubits, added two-qubit gates, additional depth and retained-shot fraction. The Clifford-only constraint is enforced at the computational boundary: qiskit-paulice synthesizes compatible checks for Clifford circuits, whereas the evaluated T-doped circuits are presented without automatic protection. The circuit inspector can show the generated Qiskit instruction sequence, including data and ancillary wires, coupling operations and readouts; the decorative graphics are kept distinct from this physics-based display.

The strategy workbench exposes a measured-benchmark versus live-run selector, clickable ordered data-qubit targets and an allocation of zero to three checks. Resource meters display ancillary-qubit use, additional entangling gates and circuit-depth expansion against the stated mission budget. An execution visualiser then presents four stages: noise exposure, check synthesis, syndrome-based acceptance or rejection, and final fidelity-retention accounting. These animations explain the recorded computation; they do not replace physical circuit traces or measured syndrome counts.

Each configuration can be evaluated through two explicitly labelled paths. In measured-result mode, the service retrieves a previous Qiskit result only when the circuit parameters, noise specification, simulation seed, shot count and protection strategy match the saved record exactly. Otherwise, it reports that the configuration has not been evaluated. In live mode, the service constructs and simulates the requested valid circuit and computes its measurements; this mode requires an operational Qiskit environment. The distinction avoids presenting estimated, interpolated or neighbouring records as if they were observations of the player's selected strategy.

Four research tasks structure the interactive sequence (Table 3). The first introduces detection by asking for a specified fidelity increase while preserving a minimum accepted-shot fraction. The second makes ancillary resources and additional gate operations scarce. The third asks the participant to match the best feasible configuration within a documented, bounded set of measured candidates, without implying a global optimum. The fourth compares unprotected Clifford and T-containing payloads and asks the user to interpret why differences in ideal output distributions do not establish noise suppression. These tasks use different scientific success conditions rather than treating a resource-feasible configuration as an automatic victory.

Research task	**Participant decision**	**Reported outcome**
Protection	Select target order and request non-zero checks	Conditional fidelity, syndromes, accepted-shot fraction
Resource limits	Allocate checks within ancilla, gate, depth and retention budgets	Feasibility verdict and measured overhead
Benchmark comparison	Find a feasible strategy within a bounded candidate set	Matched measured reference and composite-utility ratio
Non-Clifford interpretation	Compare T-doped and Clifford payloads without automatic T checks	Distinct ideal distributions and limitations of the comparison

Table 3. Scientific correspondence between the four interactive tasks, the participant's choice and the recorded physical or statistical outcome.

A preflight calculation evaluates whether the fixed mission requirements can be met by any recorded candidate under the stated budgets. For instance, the feasible set is defined by simultaneous constraints on conditional fidelity, retention, ancillary-qubit count, two-qubit gate overhead and added depth. A target fidelity above one is mathematically impossible; a target below one can still be infeasible in the tested candidate set. In either case, the interface distinguishes a physical impossibility from an unverified or unsuccessful bounded search. It

does not lower the success threshold after observing a player's choice, and a no-check circuit cannot satisfy tasks whose stated aim is to demonstrate protection.

Following execution, the gaming platform displays the baseline and accepted-outcome fidelity, the number of shots accepted and rejected, syndrome information when available, and the physical overhead of the added checks. The numerical score is defined by equation (3), and is presented alongside its two constituents rather than as a literal count of correct quantum shots. A strategy that raises conditional fidelity but discards most of the experiments can therefore receive a lower composite score. If no shot survives, conditional fidelity is explicitly undefined. The participant can compare and replay recorded decisions to examine which physical constraint or statistical trade-off explains the outcome.

The debrief provides separate plain-language and technical explanations, with syndrome summaries, replay controls and the measured composite-utility ratio. A research view further offers a gate-level Qiskit circuit inspector, a retention-versus-conditional-fidelity plot of the recorded benchmark and a provenance display for runtime settings and circuit seeds. The interface should disclose whether a displayed result is a saved measurement or a new live simulation, and a science-oriented view must retain the exact numerical values used to compute the score.

The platform is thus an interpretable interface to the same evidence underlying the circuit benchmark and a means to examine alternative resource-allocation decisions. Its game structure does not independently validate a new error-detection code, a superior selection algorithm or improved learning outcomes. Establishing educational effectiveness would require a separate controlled user study, while claims about deployability or operational speed require end-to-end performance testing. These evaluations are distinct from the quantum-simulation results reported here.

Discussion

The central finding of the reported benchmark is a separation between quality conditional on acceptance and utility across all attempted executions. Ancilla-based checks can filter out

outcomes associated with detected faults, allowing the accepted distribution to more closely match its ideal reference. Yet this improvement may not compensate for the discarded measurements and the extra noisy operations required to implement the checks. The same method can therefore be valuable for an accuracy-constrained experiment and undesirable for an aggregate-yield objective. Neither conclusion is contradictory; each follows from a different scientific requirement.

Our framework clarifies this distinction by requiring every protected-fidelity result to be reported together with its retention probability and the physical cost of obtaining it. The simple necessary condition in equation (5) makes the trade-off immediately visible. When an unprotected distribution already agrees with the ideal distribution at 96.4%, retention must exceed 96.4% for protection even to have a mathematical possibility of improving the defined *pF* score. That condition alone does not guarantee success, because the accepted fidelity must also exceed the break-even curve in equation (4). The analysis is intentionally modest: it clarifies a decision rule rather than introducing a new quantum code.

For tasks such as variational energy estimation or expectation-value measurement [19], a linear composite utility is insufficient. An error-detection strategy can reduce estimation bias while increasing variance because fewer observations remain. A more task-specific comparison would estimate the number of attempted shots required to reach a prescribed mean-squared error. For an observable with accepted-shot variance σ_s^2, conditional bias b_s, and acceptance probability p_s, a simplified independent-shot approximation is

$$\mathrm{MSE}_s(N) \approx b_s^2 + \frac{\sigma_s^2}{N p_s}$$

For a requested error tolerance ε with $\varepsilon^2 > b_s^2$, the approximate raw-shot requirement is

$$N_s(\varepsilon) \gtrsim \frac{\sigma_s^2}{p_s\left(\varepsilon^2 - b_s^2\right)}$$

These relationships show why the scientifically best protection strategy depends on the measured observable, its bias and variance, and the accepted-shot fraction; an objective built

solely from a distribution-overlap score cannot replace them. Equations (8) and (9) describe a task-specific analysis direction; they are **not** quantities evaluated in Tables 1 and 2.

The comparison also has important boundaries. First, the reported experiments use synthetic depolarizing and readout noise, with idealized single-qubit gates; actual quantum processors can exhibit heterogeneous, coherent, correlated and time-varying faults. Second, the benchmark considers small registers and a restricted family of target-qubit orders and check counts. Third, the classical distribution-fidelity measure may miss phase errors that do not affect the final measurement basis. Fourth, selection and inference on overlapping circuit instances can exaggerate apparent improvements. Fifth, the current non-Clifford evaluation does not test protected T-doped payloads. These limitations prevent extrapolation from the reported results to universal circuits, fault-tolerant hardware or a hardware-independent optimum.

The interactive gaming platform described above makes the fidelity–retention trade-off auditable at the level of an individual protection decision. Its contribution is experimental transparency and communication; its educational effects and operational performance have not yet been established by controlled evaluation.

The next experimental steps are straightforward but essential: deposit the source circuit instances and raw counts; verify package versions and the declared noise channels; complete a depth-by-noise sweep with independent seeds; provide accepted-shot uncertainty and matched paired comparisons; assess strong qiskit-paulice selection baselines; and repeat the analysis under realistic backend-derived noise and, if possible, quantum hardware. Additional implementation directions include hardware execution through Qiskit Runtime, testing learned syndrome decoders and evaluating task-specific, variance-aware decision objectives; none has yet been established by the results presented here. In conclusion, the simulations distinguish improvements in the accuracy of retained outcomes from improvements in efficiency across all attempted executions. In the evaluated regimes, the reported gain in conditional distribution fidelity did not translate into a positive composite-utility advantage. This separation provides a transparent basis for choosing when error detection is warranted, while leaving hardware generality and improved strategy learning as questions for future validation.

Code availability

The code is available at the following GitHub Link: https://github.com/naveen-joy-18/Quantum-Sentinel-An-Interactive-Gamified-Platform-for-QEC.git

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